Combining Time-Series Variation Modeling and Fuzzy Spatiotemporal Feature Fusion: A Novel Approach for Unsupervised Flood Mapping Using Dual-Polarized Sentinel-1 SAR Images

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***Abstract*— Due to the impact of climate change, the frequency of flood events has increased in recent years, which puts forward an urgent need for timely and accurate flood mapping for emergency response. As the synthetic aperture radar (SAR) enables all-time monitoring regardless of bad weather conditions, it fits far better than passive optical sensors to delineate submerged areas during flood events. However, the universal, rapid, and accurate detection of flood extent remains a challenge. Drawing inspiration from the analysis of time-series variation in representative ground objects caused by flood events, as observed in a dual-polarized SAR time series over a hydrological year, we construct a novel window-based variation model. This model can be used to capture both long-term trends and short-term fluctuations of flood features across different polarization modes. Subsequently, we introduce an unsupervised flood-mapping framework that integrates spatiotemporal flood features extracted by fuzzy-based methods. Given the distinct backscatter value of short vegetation, a flooded short vegetation (FSV) activation model is designed and performed to enhance flood-mapping accuracy in complex regions. The proposed method, tested on the 2020 East Dongting Lake flood in China, surpasses three unsupervised flood-mapping methods and two deep-learning methods in terms of quantitative evaluation and visual performance. The uncertainty of our proposed framework is tested through parameter sensitivity analyses, comparisons with flood-mapping results from other sensor images, and extensive experiments on floods at different locations and times, thereby demonstrating its effectiveness, stability, and universality.**

***Index Terms*— Flood mapping, fuzzy logic, spatiotemporal feature fusion, synthetic aperture radar (SAR), time-series variation.**

Manuscript received 1 August 2023; revised 25 September 2023; accepted 9 October 2023. Date of publication 12 October 2023; date of current version 25 October 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0715203, in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 62221002, in part by the National Natural Science Foundation of China under Grant 61901166, and in part by the Natural Science Foundation of Hunan Province of China under Grant 2022JJ40054. *(Corresponding author: Xinxin Liu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3324066

I. Introduction

With the increase of extreme precipitation events caused by climate change, floods have gradually become the most severe natural disaster in the world, causing enormous damage to people and economies [1], [2], [3], [4]. As the basis of flood control and disaster relief, flood mapping and analysis are of great significance and urgent need. Timely and precise assessment of flooded regions is the pursuit of flood mapping [5], [6]. Traditional methods rely heavily on the manual collection of flood information, which makes it easy to generate unsatisfactory flood-mapping results with insufficient timeliness or accuracy [7]. Fortunately, the increase in remote-sensing data sources provides new technical support for flood mapping. Compared with traditional methods, flood mapping based on remote sensing can realize large-scale continuous observation of the land surface and plays an important role in all the stages from flood monitoring to rescue management [8]. According to the images involved, remote-sensing flood-mapping methods can be roughly divided into two groups. The first group is characterized by using optical remote-sensing images, which draws support from the abundant spectral information and interpretability of optical images. However, as continuous rainfall is often accompanied by flood events, the quality and reliability of the optical satellite data are susceptible. The second group employs synthetic aperture radar (SAR) images to map floods, where each digital number (DN) value records the backscatter information of ground objects. Therefore, the changes in the state and characteristics of different objects can be directly reflected by the changes in backscatter values [2]. Due to the strong penetrability of SAR imaging, the impact of clouds and rainfall can be effectively reduced, which makes the SAR-based methods more powerful and widely used in flood mapping [9], [10].

To date, there have been various researches on SAR-based flood-mapping methods. Since the backscatter of water is much lower than that of other ground objects in an SAR image [11], many algorithms can quickly extract the approximate range of water bodies from an SAR image on a flood day. Thresholding methods, such as the Otsu thresholding algorithm [12] and the KI thresholding algorithm [13], are commonly employed. These techniques extract the water area

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by seeking an appropriate threshold value, which can be either local or global [14]. Another group of approaches that can quickly segment water bodies from a single SAR image is clustering-based methods, such as the k-means clustering algorithm, fuzzy clustering algorithm, and so on [15], [16], [17]. The fundamental concept of clustering algorithms is to cluster around several points and classify the objects closest to them. The clustering results are iteratively optimized with the updating of the cluster center. Different from many traditional crisp partition clustering methods, fuzzy clustering can express the membership degree of sample attribution, thereby reflecting problems more objectively and practically, which accounts for its widespread application. For instance, Boni et al. [18] designed a prototype flood-mapping system using thresholds on COSMO-SkyMed and Sentinel-1 Data. Metrikasari et al. [19] utilized Fuzzy C-Means algorithm to map flood risk in Jakarta. Cao et al. [20] built an automatic flood-monitoring system with fuzzy clustering and grading thresholds. Despite the computational efficiency of the aforementioned methods, their weak anti-interference ability against atmospheric disturbances [21], smooth bare soil with similar backscatter signal to water, interference of mountain and building shadows, and double-bounce effects caused by vegetation [22], [23], [24] are nonnegligible.

Different from the techniques of thresholding and clustering, methods based on change detection extract flooded areas by generating the difference map between images before and during a flood event, which conquers part of the limitations of thresholding [25], [26], [27], [28]. For example, Giustarini et al. [29] introduced an automatic and reliable change detection approach to map the flood in urban areas using TerraSAR-X. Amitrano et al. [30] designed a change index feedback system for flood mapping by considering the information from preflood and postflood stages. Matgen et al. [31] constructed a hybrid method combining thresholding and change detection, in which change detection alleviated the over-detection problem of thresholding. Chini et al. [32] further built a flood-mapping algorithm making use of change detection, region growing, and thresholding and proposed a hierarchical split-based approach (HSBA) to optimize the parameter selection. Nevertheless, the interference from complex backscatter value and noise easily accumulates in flood-mapping results through change detection, causing an obvious decline in extraction accuracy [31]. More importantly, there exist inner differences between different nonflood images, which means that the estimation of flooded (changed) regions deviates when different nonflood images are used as the reference in the flood extraction [33].

Faced with the uncertainty in change detection methods, an effective strategy is to obtain stable and reliable flooding information through multitemporal data. Since SAR images are almost unaffected by clouds and rainfall, it is accessible to realize regular observation of flood areas, which provides the chance to conduct the multitemporal statistical analysis [34], [35] of sequential SAR images. Recently, multitemporal statistical analysis has become another popular family of flood-mapping methods and has attracted extensive attention. Cian et al. [36] designed a normalized difference flood index by using statistical information of multitemporal SAR data to allow an easy and fast categorization of "flooded" areas. Zhao et al. [37] generated the exclusion map from the SAR time series to improve flood-mapping accuracy in the regions that were insensitive to surface changes. With the help of the Google Earth Engine (GEE) cloud computing platform, a near real-time flood identification algorithm based on multitemporal SAR statistical information was proposed in [38]. To deal with the overdetection caused by noise or strong scatterers, Shen et al. [39] developed the Radar Produced Inundation Diary (RAPID) system via statistical classification of multitemporal SAR data. In [40], the spatial and temporal distribution of floods in Bangladesh from 2014 to 2018 was analyzed, and the flood-prone paddy field areas were mapped accordingly. Due to the stability and anti-interference of the multitemporal statistics, it shows high promise in accurate flood mapping.

Despite considerable efforts, there still remain some problems that are not well solved. First, due to the complex backscatter characteristics of the radar signal, the records of ground objects under different polarization modes vary significantly. Although images from different polarization modes are all applicable for flood mapping, their varying sensitivity to different ground objects can easily lead to nonignorable errors. Second, many flood-mapping methods that use multitemporal images tend to process each image independently and then merge the results, rather than extracting temporal variation features from the image sequence. However, independent processing for each image neglects the continuous changes in image sequences caused by flood events, which may lead to false or missing extraction in steps susceptible to many factors. Furthermore, considering that SAR imaging is inevitably subject to echo interference, the direct calculation of the differences between SAR images at different times fails to capture the essential temporal variations. Thus, a concise and stable model that can conquer the echo interference and properly depict the variation needs to be proposed.

To overcome the above problems, a novel unsupervised approach that combines time-series variation modeling and fuzzy spatiotemporal feature fusion (VMSTF) is designed to accomplish fast and robust flood mapping through dual-polarized Sentinel-1 SAR time series. Based on the feature analyses of flood-related ground objects, the proposed VMSTF establishes a window-based variation model to realize a stable description of temporal changes and uses fuzzy-based methods to extract spatial and temporal features from appropriate polarized data. After fuzzy spatiotemporal feature fusion in the VMSTF framework, the flooded short vegetation (FSV) activation model is implemented to supplement flood areas that are easy to be omitted. Flood maps from perspectives of flood event analysis and flood identification are both provided by VMSTF to meet the needs of more comprehensive flood analyses. Extensive experiments demonstrate the effectiveness and stability of the proposed framework. Specifically, the contributions of this work can be summarized as follows.

1) We propose a window-based variation model based on the thorough analysis of the dual-polarization SAR time-series images of an entire hydrological year. The model

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can capture long-term trends and short-term fluctuations of flood-related features in SAR image sequences under a specific polarization mode. The introduction of the sliding time window enables the model to consider temporal correlation in the image sequence and count more representative data without the interference of many factors during water extraction.

2) We propose an unsupervised flood-mapping method that fuses temporal and spatial features to characterize floods more stably and reliably. By using fuzzy-based methods, three temporal and two spatial flood-related features are extracted, and a six-class attribute map representing land–water changes is obtained after spatiotemporal fusion.
3) Considering the inherent double-bounce effects of SAR imaging, an FSV activation model is proposed to relieve the omission problem in submerged areas. This activation model enables the framework to quickly and accurately carry out flood mapping in complex regions such as rivers and lakes with rich vegetation coverage.

The rest of this article is organized as follows. The proposed VMSTF flood-mapping framework is introduced in Section II. Experimental results and comparisons with other methods are exhibited in Section III. Section IV first analyzes the parameter sensitivity and then verifies the stability of the proposed framework in mapping flood events that occurred at different times and in different regions. Conclusions are finally drawn in Section V.

II. Methodology

This section introduces a novel flood-mapping framework based on time-series variation modeling and fuzzy spatiotemporal feature fusion. In the following context, we first analyze flood features using dual-polarization SAR time series and then introduce the variation modeling and the fuzzy-based feature extraction method, followed by spatiotemporal feature fusion and joint decision-making. As a model to supplement the flood extraction areas, the FSV activation model is finally illustrated. The overall framework is shown in Fig. 1.

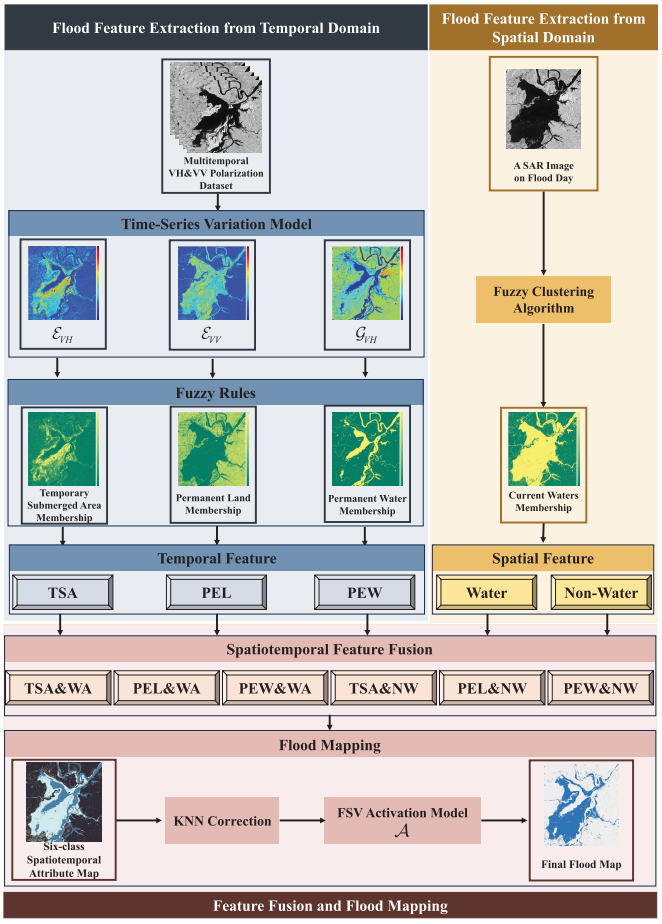


Fig. 1. Illustration of the proposed unsupervised flood-mapping framework with time-series variation modeling and fuzzy spatiotemporal feature fusion.

A. Analyses of Flood Features in SAR Time Series

According to the imaging features of SAR, the backscatter signal in water bodies is much weaker than that in land and other nonwater areas due to specular reflection [41], [42]. Namely, the water bodies are characterized by low intensity in the spatial domain that can serve as an effective feature to distinguish from the nonwater regions in the SAR images. However, the water bodies contain not only temporary open water (TOW), but also permanent water (PEW). Additionally, due to the imaging characteristics of SAR, it is vulnerable to the interference of many factors during water extraction, such as human activities or special backscattering signals of short vegetation. Thus, the key problem of accurate flood mapping is to extract floods solely from water bodies while properly dealing with the temporary interference. Obviously, using spatial features alone is not sufficient. Since flood events will lead to significant time-series variations of related ground objects, we turn our attention to analyzing the fluctuations and trends of SAR image sequences to characterize the temporal features for flood mapping. More specifically, four target ground objects, including PEW, TOW, FSV, and permanent land (PEL), are fully analyzed. Among them, TOW and FSV can be collectively referred to as temporary submerged areas (TSA).

To analyze the flood features, we plotted the average variation of backscatter signals of four target objects in a hydrological year (including flood period and nonflood period) under the dual-polarized mode in Fig. 2. It can be seen that the PEW has a relatively stable low value throughout the year in VH and VV polarizations, which means not only its reflection in the spatial domain is low, but also its short-term fluctuation and long-term trend in the temporal domain are small. This is understandable since PEW bodies, generally located at the center of lakes or rivers, rarely change during flood events. Compared with PEW, the backscatter value of PEL is much higher and possesses slight fluctuations in VH- and VV-polarized SAR time series. The main reasons for the slight fluctuations of PEL are the natural dynamics of ground objects and interference from environmental factors such as rainfall. TOW and FSV, however, vary from land or vegetation to water and finally back to land or vegetation, which can generate two opposite long-term change trends in the time series with drastic short-term fluctuations. It is

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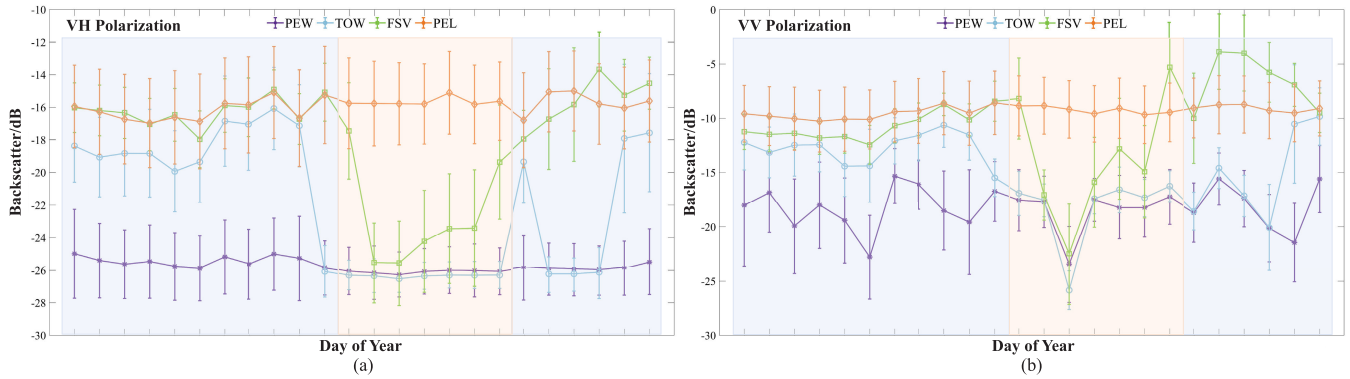


Fig. 2. Average variation of backscatter value of four target objects in a hydrological year under the dual-polarized mode (with error bars). The range of *orange* background color is flood period, while the rest is nonflood period. (a) VH polarization. (b) VV polarization (PEW, TOW, FSV, and PEL).

noteworthy that there also remain slight differences in the changing style between TOW and FSV. Specifically, the half-submerged vegetation and the smooth water surface under vegetation are prone to cause double-bounce scattering, which enhances the backscatter in flooded vegetation regions to an even higher value than that under nonflooded conditions. This phenomenon is more obvious in the line chart of VV polarization in Fig. 2. From the perspective of the temporal domain, the double-bounce mechanism makes the backscatter of FSV regions decrease later but rebound earlier than that of TOW during flood events [43], [44], [45]. Based on the different characteristics of four target ground objects in temporal and spatial domains and their latent complementarity under dual polarization analyses, we proposed an unsupervised flood-mapping method by using time-series variation modeling and fuzzy spatiotemporal feature fusing.

B. Temporal Feature Extraction of SAR Time Series via Fuzzy Rules

1) Enhanced Time-Series Variation Modeling: Floods are closely related to changes from the temporal domain. Therefore, we first construct two differential matrices D_S and D_L to separately obtain the absolute difference and difference between two images of adjacent time, written as

$$D_S(\mathbf{I}_t) = |\mathbf{I}_t - \mathbf{I}_{t-1}| \tag{1}$$

$$D_L(\mathbf{I}_t) = \mathbf{I}_t - \mathbf{I}_{t-1} \tag{2}$$

where $\mathbf{I} \in \mathbb{R}^{m\times n\times T}$ denotes sequential SAR images with the spatial size of $m\times n$ and temporal length of T, and $\mathbf{I}_t$ represents the SAR image at time t.

To stably capture the essential changes in the temporal domain, two novel temporal variation measures are designed based on the above variation matrices to characterize short-term fluctuations and long-term trends, respectively. They are calculated through the sliding window technique, expressed as

$$\mathcal{V}_S(t) = \text{median}\{D_S(\mathbf{I}_{t-r}), \ldots, D_S(\mathbf{I}_t), \ldots, D_S(\mathbf{I}_{t+r})\} \tag{3}$$

$$\mathcal{V}_L(t) = |\text{mean}\{D_L(\mathbf{I}_{t-r}), \ldots, D_L(\mathbf{I}_t), \ldots, D_L(\mathbf{I}_{t+r})\}| \tag{4}$$

where r is the radius of the sliding window in the temporal domain centered at time t. The schematic describing the calculation process is shown in Fig. 3.

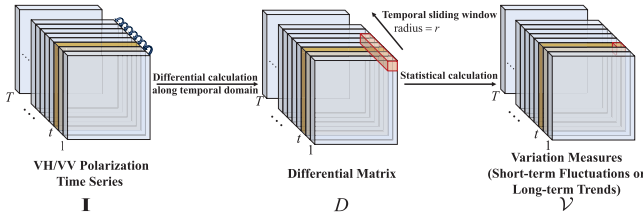


Fig. 3. Description of the calculation process of short-term fluctuations and long-term trends.

According to (1)–(4), $\mathcal{V}_S$ is a windowed short-term fluctuation measure, which takes the median value of the absolute temporal differences within a certain time range, while $\mathcal{V}_L$ works as windowed long-term trend measure, which accumulates the temporal differences within the windows. Different from D_S in $\mathcal{V}_S$, D_L in $\mathcal{V}_L$ does not take the modulus, so the long-term trend (the value of $\mathcal{V}_L$) depends on whether the temporal changes are coincident considering the temporal difference of $\mathbf{I}_t$ [the value of $D_L(\mathbf{I}_t)$] can be either positive or negative. To obtain more representative windowed short-term fluctuation, median statistics are selected in (3) because median statistics can resist more outliers and be more stable than mean statistics. As for the sliding window constructed in $\mathcal{V}_S$ and $\mathcal{V}_L$, its role is to make full use of the sudden change characteristics of flood events to build better separable models. The recommended radius of sliding window r is 3.

Through multitemporal variation measures, we obtain the feature maps of $\mathcal{V}_S$ and $\mathcal{V}_L$ where each pixel is recorded by a specific eigenvalue for short- or long-term variation. Based on the previous analysis, TSA (including TOW and FSV) possesses large values on both short-term fluctuation and long-term trend when the flood rises and recedes, whereas PEW and PEL are much more stable with low values on both short-term fluctuation and long-term trend in any periods. This observation provides us clues to build a more effective variation model for flood identification. Therefore, long-term trend measure $\mathcal{V}_S$ and short-term fluctuation measure $\mathcal{V}_L$ are combined to build the enhanced variation model $\mathcal{E}$ to increase the contrast among PEW, TSA, and PEL, given as

$$\mathcal{E} = \max_t\{\mathcal{V}_S(t) + \mathcal{V}_L(t)\} \tag{5}$$

where $\max_t$ seeks the best t to maximize the objective function. It is easy to understand that the enhanced variation model can capture the most notable change in the time series. So, model

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$\mathcal{E}$ is effective in making the TSA stand out given that the most significant changes in permanent objects are still small.

2) Dual-Polarized Feature Analysis: Although different polarizations of SAR images can be employed for flood mapping, each polarization can cause greater classification errors in certain objects [46], [47]. For example, cross-polarized data (VH and HV) are insensitive to waves on the water surface, but vulnerable to low backscatter values, misclassifying the land into flooded regions. On the contrary, VV-polarized data can capture the roughness of surfaces but is susceptible to water surface under wind conditions. Since the advantages and disadvantages of VH and VV polarizations are complementary, dual-polarization information (from IW-mode Sentinel-1 data) is fully utilized in the proposed flood-mapping framework. Specifically, VH- and VV-polarized data are fed into the enhanced variation model $\mathcal{E}$, generating maps of $\mathcal{E}_{\mathrm{VH}}$ and $\mathcal{E}_{\mathrm{VV}}$, to separately highlight TSA and PEL.

The enhanced variation model seems to be competent to highlight the PEW, but it is vulnerable to passing ships and human activities when seeking the most significant changes. To ensure a reliable extraction, this work then draws support from the SAR time series during nonflood periods with the temporal length of l denoted by $\mathbf{R} \in \mathbb{R}^{m\times n\times l}$. Since $\mathbf{R}$ is not affected by floods, it can be used as the reference image sequence to reflect the general conditions of the study area. Concretely, we calculate the median value of the reference set $\mathbf{R}$ along the temporal direction as follows:

$$\mathcal{G} = \mathrm{median}\{\mathbf{R}\}. \tag{6}$$

Obviously, $\mathcal{G}$ represents the general state of the ground surface, which is not affected by exceptional circumstances or environment in a single SAR image. Based on the aforementioned prior knowledge that VH-polarized data has advantages in characterizing water areas, VH-polarized reference sequences are chosen to generate a map of $\mathcal{G}_{\mathrm{VH}}$ to extract PEW.

3) Fuzzy Rule Establishment: To efficiently obtain the distribution of TSA, PEL, and PEW from the temporal domain, maps of $\mathcal{E}_{\mathrm{VH}}$, $\mathcal{E}_{\mathrm{VV}}$, and $\mathcal{G}_{\mathrm{VH}}$ are further transformed to describe the membership degrees of objects of interest based on fuzzy logic. Before being fuzzified, $\mathcal{E}_{\mathrm{VH}}$, $\mathcal{E}_{\mathrm{VV}}$, and $\mathcal{G}_{\mathrm{VH}}$ are normalized between [0, 1]. Therefore, at each pixel location, three membership degrees can be acquired corresponding to different objects. The parameter setting of the fuzzy set is shown in Table I. The Z-shaped and S-shaped membership functions are applied in the proposed flood-mapping framework. Their definitions are given as follows:

$$f(x;a,b)_{\mathrm{z-shaped}} = \begin{cases} 1, & x \le a \\ 1-2\left(\dfrac{x-a}{b-a}\right)^2, & a \le x \le \dfrac{a+b}{2} \\ 2\left(\dfrac{x-b}{b-a}\right)^2, & \dfrac{a+b}{2} \le x \le b \\ 0, & x \ge b \end{cases} \tag{7}$$

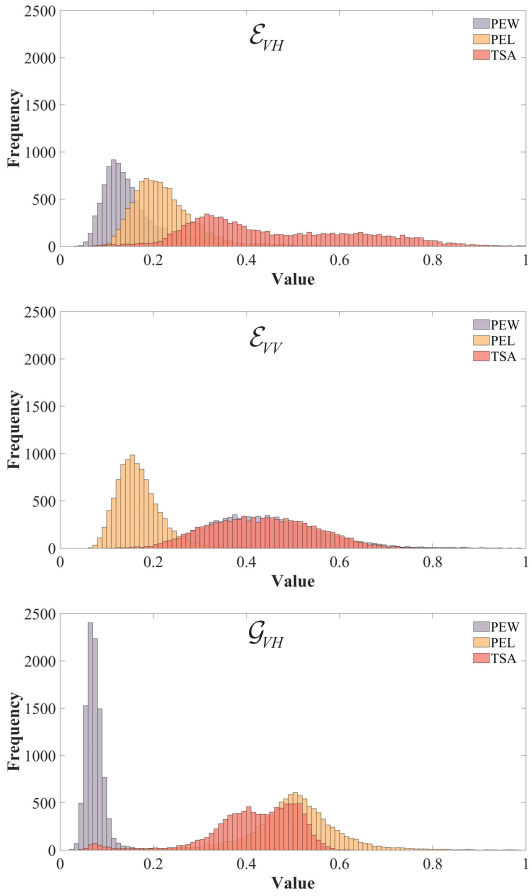


Fig. 4. Histogram distribution of PEL, PEW, and TSA on normalized maps of $\mathcal{E}_{\mathrm{VH}}$, $\mathcal{E}_{\mathrm{VV}}$, and $\mathcal{G}_{\mathrm{VH}}$ (PEW, PEL, and TSA).

$$f(x;a,b)_{\mathrm{s-shaped}} = \begin{cases} 0, & x \le a \\ 2\left(\dfrac{x-a}{b-a}\right)^2, & a \le x \le \dfrac{a+b}{2} \\ 1-2(\dfrac{x-b}{b-a})^2, & \dfrac{a+b}{2} \le x \le b \\ 1, & x \ge b \end{cases} \tag{8}$$

where x represents the feature value of each pixel in normalized maps of $\mathcal{E}_{\mathrm{VH}}$, $\mathcal{E}_{\mathrm{VV}}$, and $\mathcal{G}_{\mathrm{VH}}$, a and b are membership function parameters, with a and b separately defining shoulder and foot of the z-shaped membership function and the opposite of the s-shaped membership function. The parameters of the membership functions can be determined by analyzing the histogram distribution of TSA, PEL, and PEW on normalized maps of $\mathcal{E}_{\mathrm{VH}}$, $\mathcal{E}_{\mathrm{VV}}$, and $\mathcal{G}_{\mathrm{VH}}$. By manually selecting some typical areas of target objects as an example, the histogram distribution is drawn and shown in Fig. 4. It can be seen that each of the target objects can be clearly separated on a certain map, which verifies the separability of the proposed model and simplifies the choices of parameters.

C. Spatial Feature Extraction of Current SAR Image via Fuzzy Clustering

Through the extraction of multitemporal flood features, we can only know the affected areas in the whole flood

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TABLE I
PARAMETER SETTING OF THE FUZZY SETS RELEVANT TO TSA, PEL, AND PEW

Map	Target object	Membership function	a	b
$\mathcal{E}_{VH}$	Temporary Submerged Area	S-shaped	0.2	0.7
$\mathcal{E}_{VV}$	Permanent Land	Z-shaped	0.1	0.3
$\mathcal{G}_{VH}$	Permanent Water	Z-shaped	0.1	0.2

event, but cannot get the flooded state at different times. To achieve accurate and instantaneous flood mapping, this work then incorporates spatial instantaneous flood features into the proposed framework.

To acquire spatial instantaneous features corresponding to water and nonwater regions from a single SAR flood image, the group of minimization-based clustering algorithm is a good choice for stable performance [48]. Among this group, the fuzzy C-means (FCM) clustering algorithm [16] is one of the most popular methods because of its theoretical completeness and objectivity. FCM is a soft algorithm that uses membership to describe the belonging degree of each pixel and improves the clustering performance at the cost of increasing iterations [49]. Accordingly, FCM can obtain the membership of spatial features, which is just suitable to integrate the membership of multitemporal features to complete the spatiotemporal feature fusion. The objective function of FCM is given as follows:

$$J_m = \sum_{i=1}^{N}\sum_{k=1}^{C}\left(\mu_t^{ik}\right)^m \left\| I_t^i - c_t^k \right\|^2 \tag{9}$$

where N and C separately represent the number of pixels, and the number of clusters in I_t, $m > 1$ is the fuzzy partition matrix exponent for controlling the degree of fuzzy overlap. I_t^i denotes the ith pixel in I_t, c_t^k denotes the center of the kth cluster, and μ_t^{ik} gives the membership degree of I_t^i in the kth cluster. By minimizing J_m, μ_t^{ik}, and c_t^k can be updated iteratively until the stopping condition is satisfied [50].

D. Spatiotemporal Feature Fusion and FSV Activation

The basic idea is to identify the regions of instantaneous flood by simultaneously considering the features of PEW, TSA, and PEL in the temporal domain (from sequential SAR images) and the features of water and nonwater in the spatial domain (from the SAR image on a flood day) via spatiotemporal feature fusion and joint decision-making.

1) Feature Fusion and Decision-Making: From the perspective of the spatial domain, each pixel owns a membership degree of "Water" (μ_{water}) within [0,1] and a membership degree of "Nonwater" equaling to 1-μ_{water}, while in temporal domain, it owns three membership degrees of "TSA" (μ_{TSA}), "PEW" (μ_{PEW}), and "PEL" (μ_{PEL}). By synthesizing the fuzzy membership vectors in both spatial and temporal domains, each pixel can be represented via a fuzzy relationship matrix. All possible connection relationships are displayed in Table II. Defuzzification is then performed based on the maximum membership principle [51] to acquire a spatiotemporal synthetic six-class attribute map. Considering the natural continuity of ground objects, isolated pixels caused by spatial or temporal noise need to be corrected in the attribute map.

TABLE II
RULES OF SPATIOTEMPORAL FEATURE FUSION BASED ON LOGICAL ANALYSIS

Temporal feature	Spatial feature	Fusion rule
Temporary Submerged Area	Water	Flood
Permanent Water	Water	Non-Flood
Permanent Land	Water	Flood
Temporary Submerged Area	Non-Water	Flood/Non-Flood
Permanent Water	Non-Water	Non-Flood
Permanent Land	Non-Water	Non-Flood

To tackle this problem and increase the connection between pixels, the K-nearest neighbors (KNNs) method [52] is used in the proposed framework. Specifically, each isolated pixel at the center of the $k \times k$ window will be classified into the category with the highest proportion within its window. The recommended value of k is 9.

Furthermore, logical analysis is performed on the spatiotemporal synthetic attribute map to determine the flooded state as being "Flood" or "Nonflood." Concretely, TSA&WA and PEL&WA are determined as "Flood," while PEW&WA, PEW&NW, and PEL&NW are classified as "Nonflood." It is worth noting that the attribute of TSA&NW needs further analysis in the next part due to the special reflection mechanism of semisubmerged short vegetation. Basic spatiotemporal feature fusion rules are given in Table II.

2) FSV Activation Model for Flood-Mapping Supplement: The deterministic mapping of Flood/Nonflood can be ensured in most fusion cases, except when the pixel has the temporal feature of "TSA" and spatial feature of "Nonwater" (short for TSA&NW). Specifically, pixels belonging to TSA&NW can be marked as "Flood" for half-submerged short vegetation or "Nonflood" for unsubmerged land. Uncertainty mainly comes from the double-bounce effect caused by half-submerged vegetation. This effect increases the backscatter of half-submerged vegetation regions to an abnormally high value, which easily leads to insufficient extraction of flood areas. To supplement flood extraction areas, a novel activation model $\mathcal{A}$ is designed by capturing the half-submerged transient state of short vegetation, given as follows:

$$\mathcal{A} = W \circ \max\left\{\mathbf{I}_t - \max_t\{\mathbf{R}\}, 0\right\} \tag{10}$$

where $\circ$ denotes element-wise multiplication, $\max_t\{\mathbf{R}\}$ calculates the maximum value along the temporal direction of the nonflood reference SAR image sequence, and W is a mask matrix with values $W^i = 1$ for ith pixel belonging to TSA&NW and $W^i = 0$ for the opposite. Clearly, the FSV activation model $\mathcal{A}$ can well capture the double-bounce enhanced backscatter of half-submerged vegetations and generate an activation map to highlight the regions with higher backscatter values than that under nonflooded conditions. Thus, the pixels with values $\mathcal{A}^i = 1$ are regarded as half-submerged vegetation and marked as "Flood," while the rest of pixels within TSA&NW are marked as "Nonflood." Given that VV-polarized data acquire higher signal responses under double-bounce conditions than VH-polarized data [47], VV-polarized data is recommended to perform the proposed FSV activation model.

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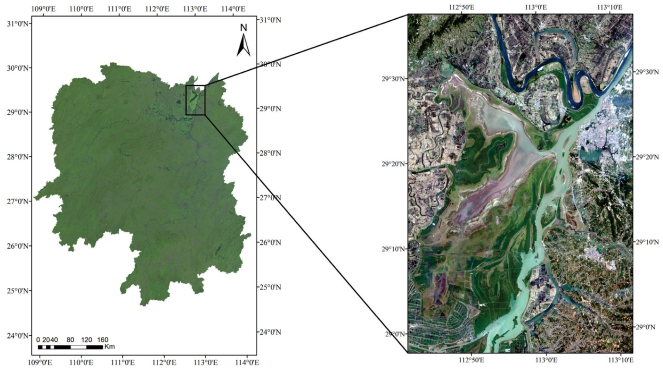


Fig. 5. Location of the study area.

To clarify the proposed unsupervised flood-mapping method, the detailed steps are summarized in the Algorithm 1.

Algorithm 1 Proposed Unsupervised Flood-Mapping Method

Input: Dual-polarized SAR time series **I**.

1: Calculate differential matrices D_L and D_S.
2: Calculate novel temporal variation measures $\mathcal{V}_L$ and $\mathcal{V}_S$.
3: Build the enhanced time-series variation model of $\mathcal{E}_{VH}$, $\mathcal{E}_{VV}$, and $\mathcal{G}_{VH}$ from dual-polarized SAR time series **I**.
4: Calculate the membership degrees in the temporal domain by performing fuzzy rules on the enhanced time-series variation model.
5: Calculate the membership degrees in the spatial domain by the FCM algorithm.
6: Fuse the temporal and spatial features and correct isolated pixels using the KNN method to obtain the six-class attribute map.
7: Perform the logical analysis and the FSV activation model $\mathcal{A}_{VV}$ on pre-extracted flood map.

Output: Final flood map **F**

III. Experiments

A. Study Area

As the second largest freshwater lake in China, Dongting Lake, located in the middle reaches of the Yangtze River, belongs to a subtropical monsoon climate with humid summer and dry winter. The average annual precipitation ranges from 1400 to 1500 mm and approaching 70% of precipitation occurs during the rainy season. Due to the continuous heavy rain concentrated in the rainy season, Dongting Lake is vulnerable to floods between June and August each year. To evaluate the proposed method, East Dongting Lake, the largest lake in the Dongting Lake system, was selected as the study area. It covers land from 112°25′ to 113°15′E and 28°50′ to 29°50′N, as shown in Fig. 5. With climate change, the frequency of flood events has increased in recent years, causing great threats to the lives and property of people. The last flood occurred between July and August 2020. At that time, the continuous heavy rain kept the water level of East Dongting Lake continuously higher than the warning value, generating the longest flood duration since 2000.

B. Data and Preprocessing

Sentinel-1 is the first satellite constellation of the Copernicus Program run by the European Union (EU) and the European Space Agency (ESA). It consists of Sentinel-1A and Sentinel-1B SAR satellites, launched on April 3, 2014, and April 22, 2016, respectively. Each Sentinel-1 satellite orbits the globe every 12 days and provides freely available data to the public. More detailed information on satellite acquisition parameters can be found in the user guide (https://sentinel.esa.int/web/sentinel/user-guides/sentinel-1-sar). For our study, the data were acquired from Sentinel-1 C-band (5.407 GHz) instrument in the mode of interferometric wide (IW) swath. The Level-1 ground range detected (GRD) products in dual-polarization (VH and VV) with a spatial resolution of 10 m were used in the experiments. To monitor the flood event of East Dongting Lake in 2020 and analyze the dynamic changes of the floodwater, 24 scenes of available dual-polarized Sentinel-1 SAR images of East Dongting Lake, 2020, were selected. Specifically, images during the flood period included seven scenes from July 1 to September 11, while the rest 17 scenes were regarded as nonflood period images.

At the same time, ground-truth data is needed for evaluation. However, it is difficult to find an independent dataset that records the dynamic changes of the floodwater to verify the flood identification accuracy of the proposed method at different times. Therefore, we manually delineated the flood and nonflood areas. 10-m resolution maps of Finer Resolution Observation and Monitoring of Global Land Cover (FROM-GLC) [53] were also employed to help determine the unchanged nonflood regions. The manually interpreted ground truth is exhibited in the fourth row of Fig. 6.

The preprocessing procedure of SAR data included the standard steps as follows: 1) orbit correction; 2) thermal noise removal; 3) radiometric calibration; and 4) terrain correction. Besides, the Lee sigma filter with a window size of 7 × 7 was performed to reduce the impact of speckle noise since speckle noise is a granular interference that inherently exists in SAR images. Finally, backscatter intensity (σ^{int}) was converted into backscatter coefficient (σ^0) via log scaling using (11). This is because the use of logarithmic (i.e., dB) units can help to amplify the differences between small backscatter signals. All the steps mentioned above were implemented on the Sentinels Application Platform (SNAP) software platform

$$\sigma^0 = 10 \times \log \sigma^{\text{int}}. \tag{11}$$

C. Flood-Mapping Performance of VMSTF

Fig. 6 exhibits the flood-mapping performance of VMSTF at different times, which clearly shows the spatial extent, the progression, and the recession of flood throughout the target flood event. Specifically, the submerged area grew rapidly from July 1, 2020, and remained at the peak flooding stage for more than a month. The obvious recession can be recognized on September 11, 2020. All the results are consistent with the raw SAR GRD product and the ground-truth data, as displayed in the first and fourth rows of Fig. 6, respectively. Due to

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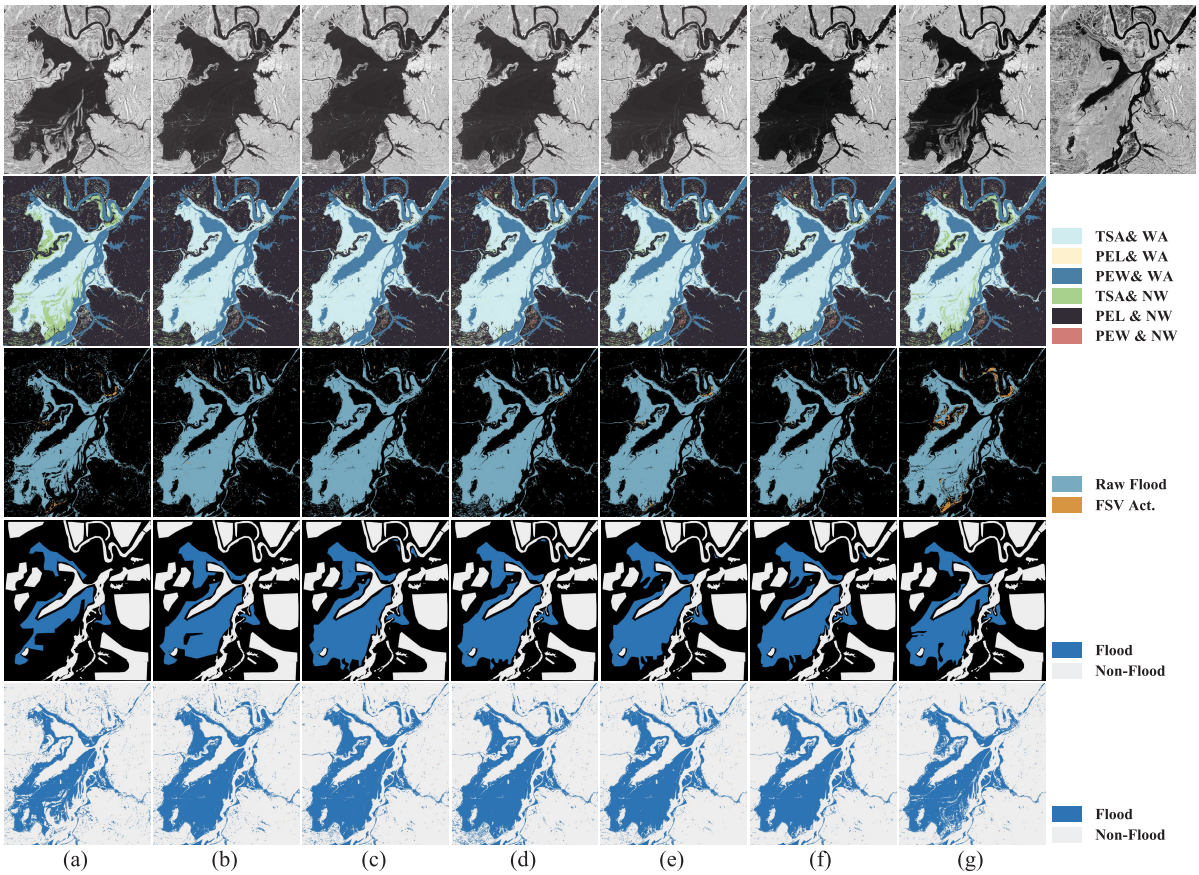


Fig. 6. Flood-mapping performance of VMSTF at each step during the flood period in 2020 in East Dongting Lake. From top to bottom: Sentinel-1 SAR images (shown in the VH channel, the last one is a nonflood image for reference), synthetic six-class attribute maps, FSV activation, ground truth, and final flood maps. (a) July 1. (b) July 13. (c) July 25. (d) August 6. (e) August 18. (f) August 30. (g) September 11 (WA: Water; NW: Nonwater).

the use of dual-domain feature modeling, the spatiotemporal characteristics of flood can be easily obtained through the synthetic six-class attribute map and then combined with logical analysis to extract the submerged areas only related to the flood. Therefore, the flood-mapping results have the advantage of being unaffected by PEW bodies, indicating good practicability of the proposed method.

Activation regions of FSV are also given in Fig. 6. As shown in the third row, FSVs concentrate mainly at the beginning and end of the flood event, but rarely in the middle stage of the flood. It is reasonable since short vegetation is more likely half-submerged when flood rises and recedes. As for small regions of half-submerged vegetation appearing on the water's edge between August 18 and September 11, this is because the spatial extent of flood fluctuates slightly during the maximum flood period.

D. Comparative Experiments

We compared our method with three unsupervised flood-mapping methods and two deep-learning-based semantic segmentation methods. Unsupervised flood-mapping methods are Otsu's thresholding method [12], the multitemporal normalized difference flood index (NDFI) method [36], and the fuzzy-based change detection (FCD) method [30]. Deep-learning methods include the 3-D convolutional neural network (3DCNN) [54] and the random patched network (RPNET) [55]. The Otsu method selects a threshold through a histogram of the image and divides images into two categories based on this threshold. The NDFI method extracts flooded areas by constructing two formulas to highlight the submerged area and semisubmerged vegetation. The FCD algorithm establishes fuzzy rules and change detection index based on preevent and postevent images to achieve flood mapping. All parameters used in the comparison methods were selected as the recommended parameters given in the corresponding articles. From the angle of a fair comparison, we made some adjustments to the implementation of Otsu and FCD methods, so that they can extract the areas only relevant to floods. In detail, the SAR image on May 26, 2020, was selected to reflect the preevent normal state for Otsu's thresholding and FCD method. Since NDFI is designed to highlight the TSA, no adjustment was made when performing this method. 3DCNN and RPNET are both supervised deep-learning semantic segmentation methods requiring a small number of labeled samples as input. The difference is that the former performs 3-D convolutions on the spatial and temporal components, while the latter utilizes randomly obtained patches from the original image as the convolution kernel. To compare with the unsupervised methods as fairly as possible, we fed the two deep-learning networks with the same input as the proposed VMSTF containing the dual-polarized

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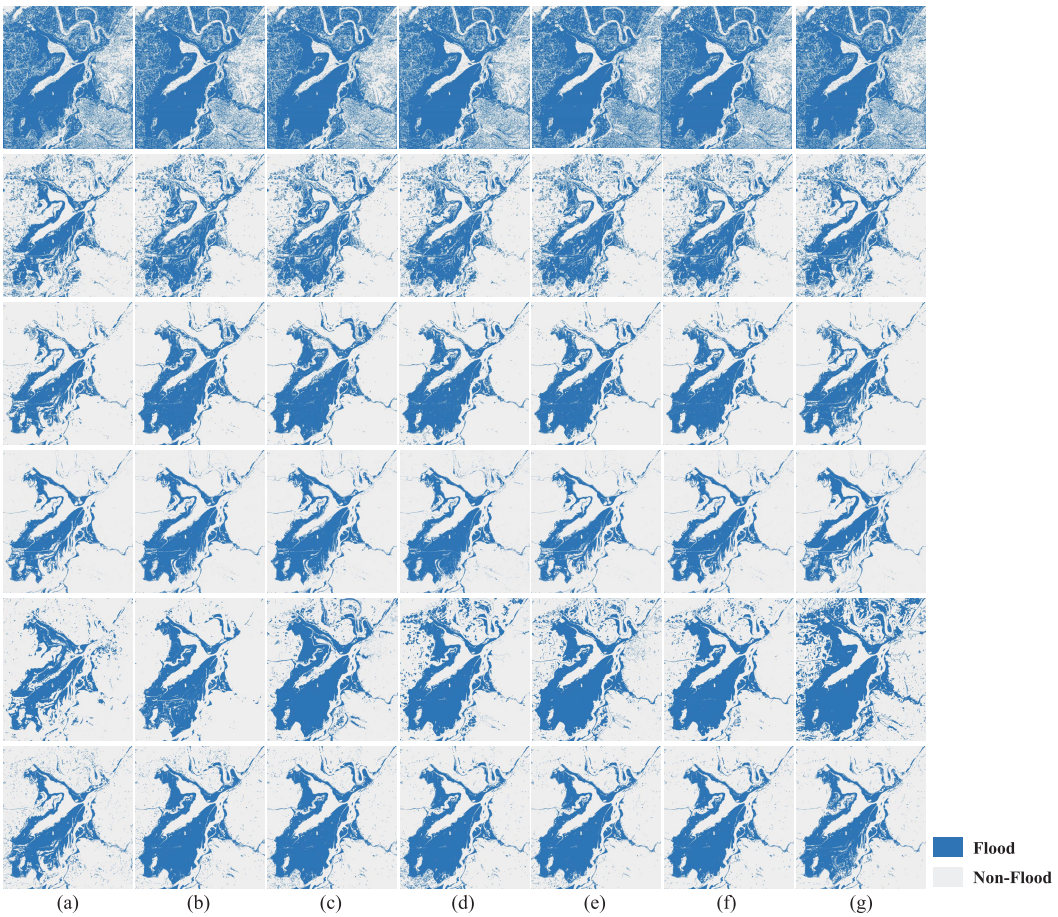


Fig. 7. Overall flood-mapping performance of different methods during the flood period in 2020 in East Dongting Lake. From top to bottom: Otsu [12], NDFI [36], FCD [30], 3DCNN [54], RPNET [55], and VMSTF. (a) July 1. (b) July 13. (c) July 25. (d) August 6. (e) August 18. (f) August 30. (g) September 11.

nonflood SAR image set and a SAR image on a flood day. They were both trained with 20 randomly selected samples including ten flood and ten nonflood labeled pixels. The rest manually labeled ground-truth pixels were used for testing. A quantitative assessment was conducted based on three full-reference indices including overall accuracy (OA), Kappa coefficient, and $F1$ score. Among them, OA measured the ratio of correctly classified samples to total test samples. Kappa coefficient evaluated how better the classification performed than completely random values, and the $F1$ score assessed the classification performance by considering both commission and omission errors. In general, the better classification results are the higher values of OA, Kappa coefficient, and $F1$ score can be obtained. Corresponding quantitative results are shown in Table III and their visual performance is shown in Figs. 7 and 8.

As is shown in Table III, the proposed flood-mapping framework can reach average values of 96.31% in OA, 94.91% in $F1$ score, and 0.92 in Kappa coefficient and can simultaneously maintain a relatively stable performance during the flood event even when the flood extent changes significantly. According to the flood-mapping results in Fig. 7, Otsu can only roughly extract the flooded area based on the difference between a single flood image and the reference image by a threshold, which cannot dig out the information in the time series. NDFI utilizes the reference image time series to distinguish flood and nonflood regions. However, there remain many block-shaped errors in each flood-mapping result

TABLE III
QUANTITATIVE ASSESSMENT OF FLOOD-MAPPING RESULTS IN 2020 IN EAST DONGTING LAKE (THE BEST RESULTS ARE SHOWN IN BOLD)

Date	Index	Otsu	NDFI	FCD	3DCNN	RPNET	VMSTF
Jul.1	F1	0.7078	0.8195	0.8977	0.8922	0.8085	**0.9073**
	Kappa	0.5485	0.7506	0.8632	0.8594	0.7421	**0.8739**
	OA	0.7788	0.8995	0.9479	0.9496	0.9010	**0.9507**
Jul.13	F1	0.7996	0.8122	0.9374	0.8580	0.9327	**0.9484**
	Kappa	0.6245	0.7029	0.8992	0.7931	0.8964	**0.9163**
	OA	0.8084	0.8616	0.9524	0.9085	0.9527	**0.9604**
Jul.25	F1	0.7949	0.8038	0.9368	0.9139	0.9157	**0.9502**
	Kappa	0.6204	0.6925	0.8990	0.8631	0.8540	**0.9206**
	OA	0.8063	0.8577	0.9526	0.9357	0.9285	**0.9628**
Aug.6	F1	0.7964	0.8216	0.9353	0.8709	0.9277	**0.9495**
	Kappa	0.6314	0.7201	0.8992	0.8008	0.8795	**0.9207**
	OA	0.8115	0.8706	0.9535	0.9079	0.9422	**0.9633**
Aug.18	F1	0.7437	0.828	0.9317	0.8606	0.9407	**0.9493**
	Kappa	0.5265	0.7323	0.8951	0.7900	0.9036	**0.9213**
	OA	0.7511	0.8769	0.9522	0.9050	0.9544	**0.9639**
Aug.30	F1	0.7868	0.8428	0.9591	0.9077	0.9535	**0.9808**
	Kappa	0.5504	0.7091	0.9201	0.8556	0.9246	**0.9621**
	OA	0.7665	0.8627	0.9601	0.9331	0.9644	**0.9811**
Sep.11	F1	0.7880	0.8526	0.9133	0.8671	0.8720	**0.9582**
	Kappa	0.5552	0.7056	0.8382	0.8050	0.7910	**0.9195**
	OA	0.7712	0.8499	0.9192	0.9141	0.9002	**0.9598**

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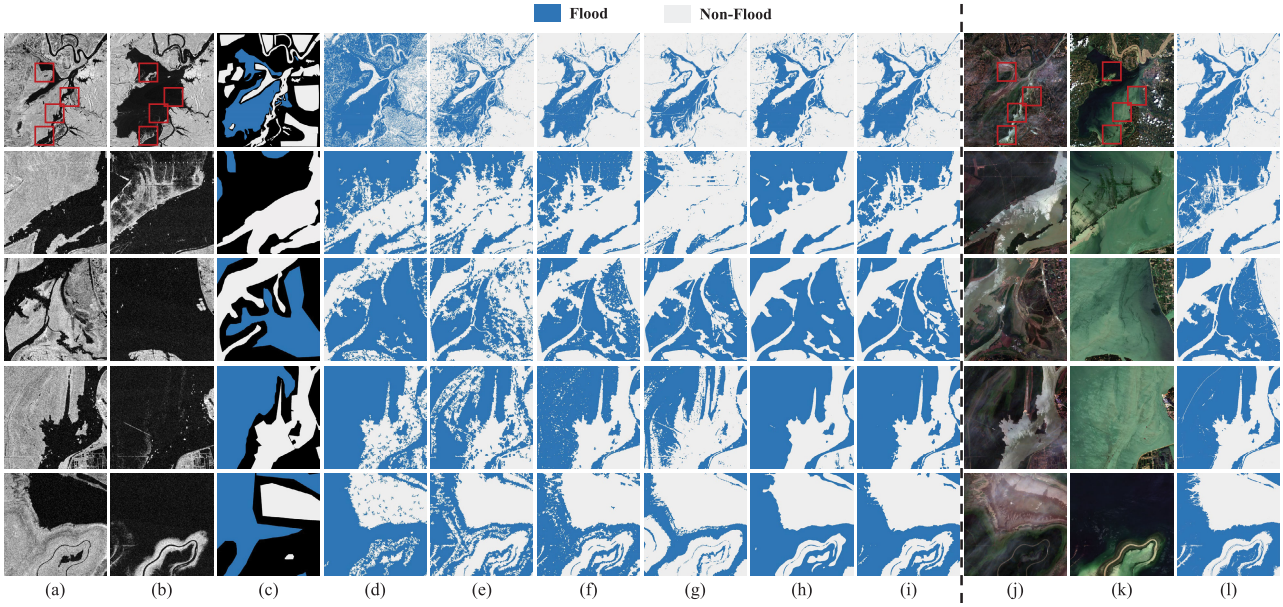


Fig. 8. Visual comparisons of detailed flood-mapping results on August 30, 2020, in East Dongting Lake. From top to bottom are complete extraction results of the flood and four closeups of the red rectangle marked regions. The left side of the dashed line shows the SAR images and their flood-mapping results, while the right side displays optical images and their flood-mapping results for reference. (a) Sentinel-1 SAR images on a nonflood day (May 26, 2020, shown in the VH channel). (b) Sentinel-1 SAR images on a flood day (August 30, 2020, shown in the VH channel). (c) Ground truth. (d) Flood-mapping result of Otsu. (e) Flood-mapping result of NDFI. (f) Flood-mapping result of FCD. (g) Flood-mapping result of 3DCNN. (h) Flood-mapping result of RPNET. (i) Flood-mapping result of VMSTF. (j) Sentinel-2 optical images on a nonflood day (May 12, 2020, shown in the RGB channel). (k) Sentinel-2 optical images on a flood day (August 30, 2020, shown in the RGB channel). (l) Flood-mapping result of the NDWI index.

due to its insufficient utilization of temporal information. For example, some paddy fields are wrongly extracted as flooded areas. FCD extracts the flood inundation range through the fuzzy semantic attributes of the change index calculated from a single flood period image and a reference nonflood image. Without reference to time-series information during the nonflood period, the fuzzy semantic attributes related to changes at the boundary between flood and nonflood regions will vary on different flood days, which affects the consistency of mapping results in the whole flood event. Both 3DCNN and RPNET are fed with temporal reference images for training, thus they can learn relatively comprehensive and stable time-series features of ground objects. However, in the case of small-sample learning, it is difficult for 3DCNN and RPNET to achieve optimal segmentation performance in complex flood events. As shown in Fig. 7, 3DCNN fails to capture the changes in different flood days during the flood period, while RPNET does not perform well in mapping stability. Among these methods, our unsupervised VMSTF has the most precise, consistent, and stable flood-mapping performance due to the combination of time-series variation modeling and fuzzy spatiotemporal feature fusion. In addition, the proposed VMSTF outperforms these compared methods with the highest OA, Kappa coefficient, and $F1$ score on each date, which also confirms the effectiveness and stability of the proposed method.

More detailed visual comparisons of flood-mapping results and the closeups of four typical areas on August 30, 2020, are displayed in Fig. 8. To analyze the effectiveness of methods from the perspective of data uncertainty [56], the Sentinel-2 optical images and the corresponding flood-mapping results are also displayed on the right side of the dashed line in Fig. 8. Specifically, the water areas of optical images are first extracted through the commonly used normalized difference water index (NDWI) [57] and Otsu's thresholding method. Then, the flood-mapping results are obtained by subtracting the water areas during the nonflood period from those during the flood period. According to the preevent and postevent GRD products and the extraction results, Otsu mistakes many nonflood areas for flood areas, while NDFI does the opposite. Many factors in SAR images in the PEW area such as passing ships or surface waves may have a greater interference on their results. The overall visual performance of FCD is better than Otsu and NDFI, but there are still many discontinuous holes in the corresponding close-ups. Perhaps, due to insufficient training samples, significant errors can be observed in the results of 3DCNN. As a patch-based convolutional neural network, RPNET performs better than 3DCNN in the same case of very few training samples. However, in its enlarged closeups, the detail extraction ability of the RPNET is still limited. Among these methods, VMSTF has the fewest holes and the most continuous extraction performance in the closeups in Fig. 8, which well reflects that our proposed method can accurately distinguish between flood and nonflood areas by combining time-series variation modeling and fuzzy spatiotemporal feature fusion. Moreover, the flood-mapping results of the proposed method are highly consistent with those of the reference optical images, which also indicates the reliability of our method.

IV. Discussion

A. Parameter Sensitivity

In the proposed flood-mapping framework, it is important to accurately characterize the spatiotemporal flood features from multitemporal SAR data. Hence, the influence of the sliding

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TABLE IV
SENSITIVITY ANALYSIS OF SLIDING WINDOW SIZE IN VARIATION MODELING (THE BEST RESULTS ARE SHOWN IN BOLD)

		Jul.1	Jul.13	Jul.25	Aug.6	Aug.18	Aug.30	Sep.11	AVG
F1	w=5	0.9018	0.9449	0.9498	**0.9504**	0.9487	0.9707	0.9286	0.9421
	w=6	0.9013	0.9447	0.9493	0.9489	0.9483	0.9702	0.9001	0.9375
	w=7	**0.9073**	0.9484	**0.9502**	0.9495	0.9493	**0.9808**	**0.9582**	**0.9491**
	w=8	0.9022	0.9452	0.9501	0.9492	0.9491	0.9711	0.8993	0.9380
	w=9	0.9025	**0.9486**	0.9502	0.9491	**0.9494**	0.9712	0.9294	0.9429
Kappa	w=5	0.8724	0.9141	0.9204	0.9203	0.9192	0.9614	0.8918	0.9142
	w=6	0.8718	0.9138	0.9201	0.9207	0.9185	0.9607	0.8752	0.9115
	w=7	**0.8739**	**0.9163**	**0.9206**	0.9207	**0.9213**	0.9621	**0.9195**	**0.9192**
	w=8	0.8728	0.9144	0.9195	**0.9209**	0.9197	0.9619	0.8744	0.9119
	w=9	0.8731	0.9161	0.9205	0.9206	0.9201	**0.9622**	0.8929	0.9151
OA	w=5	0.9505	0.9604	0.9624	0.9627	0.9625	0.9732	0.9512	0.9604
	w=6	0.9493	0.9603	0.9622	0.9624	0.9622	0.9729	0.9339	0.9576
	w=7	0.9507	0.9604	**0.9628**	**0.9633**	**0.9639**	**0.9811**	**0.9598**	**0.9631**
	w=8	0.9502	0.9605	0.9627	0.9629	0.9627	0.9734	0.9335	0.9580
	w=9	**0.9508**	**0.9606**	0.9626	0.9631	0.9629	0.9736	0.9517	0.9608

TABLE V
SENSITIVITY ANALYSIS OF FUZZY SET PARAMETERS (THE BEST RESULTS ARE SHOWN IN BOLD)

	-10%			-5%			**0**			5%			10%		
	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa	OA
Jul.1	0.9075	0.8737	0.9504	**0.9077**	**0.8743**	**0.9508**	0.9073	0.8739	0.9507	0.9049	0.8711	0.9498	0.9027	0.8692	0.9488
Jul.13	0.9448	0.9137	0.9602	**0.9506**	0.9136	0.9602	0.9484	**0.9163**	**0.9604**	0.9458	0.9148	0.9601	0.9479	0.9161	0.9604
Jul.25	**0.9517**	**0.9212**	0.9624	0.9509	0.9207	0.9622	0.9502	0.9206	**0.9628**	0.9482	0.9177	0.9609	0.9485	0.9165	0.9604
Aug.6	**0.9518**	**0.9220**	0.9630	0.9491	0.9207	0.9624	0.9495	0.9207	**0.9633**	0.9492	0.9175	0.9617	0.9479	0.9156	0.9606
Aug.18	0.9493	0.9199	0.9627	0.9483	0.9185	0.9622	**0.9493**	**0.9213**	**0.9639**	0.9467	0.9175	0.9622	0.9450	0.9149	0.9610
Aug.30	0.9762	0.9571	0.9785	0.9752	0.9606	0.9728	**0.9808**	**0.9621**	**0.9811**	0.9748	0.9574	0.9784	0.9767	0.9554	0.9785
Sep.11	0.9575	0.9028	0.9536	0.9562	0.9114	0.9561	**0.9582**	**0.9195**	**0.9598**	0.9565	0.9121	0.9501	0.9557	0.9077	0.9495
AVG	0.9484	0.9157	0.9616	0.9483	0.9171	0.9609	**0.9491**	**0.9192**	**0.9631**	0.9466	0.9154	0.9605	0.9464	0.9136	0.9599

window size in the variation modeling and the parameters in fuzzy sets should not be ignored.

The size of the sliding window represents the observation scale on the time series, which is used to describe the short-term fluctuations and long-term trends in the temporal-domain variation modeling. Given that the revisit period of Sentinel-1 data is 12 days, a too-small window size is prone to obtain unrepresentative change information due to random errors, whereas a too-large window size makes it difficult to capture long-term changes caused by floods. Although odd numbers are preferred window sizes, even numbers were also considered in the sensitivity analysis. By testing window sizes w from 5 to 9, its effects on the flood-mapping performance are shown in Table IV. It is clear that the optimum window size is 7 (i.e., the window radius $r = 3$) since it achieves the best average performance in the whole flood events under the evaluation of all indices. Also note that the differences in flood-mapping accuracy under different window sizes are very small on most dates, which reflects the stability of the proposed VMSTF framework.

For the parameters in fuzzy sets, we analyzed the influence of their changes on the accuracy of flood mapping. Specifically, the tested parameters were increased and decreased by 5% and 10% on the basis of the recommended values (see Table I). The corresponding results are reported in Table V. As we expected, the best overall performance appears when using the recommended parameters. With the variation of parameters, the average evaluation values range slightly within $[0.9464, 0.9491]$ for $F1$, $[0.9136, 0.9192]$ for Kappa, and $[0.9599, 0.9631]$ for OA. It shows that the performance of VMSTF is insensitive to the changes of parameters in fuzzy sets.

In fact, due to the complexity of different flood events, we cannot guarantee that the recommended parameters are always the best. Fortunately, the parameters in the proposed method are insensitive, which makes parameter adjustment relatively simple and convenient in real applications.

B. Transferability Across Flood Locations

To verify the robustness of the proposed VMSTF method, two other flood cases occurring around different lakes were selected for testing. Detailed information is introduced as follows.

1) Poyang Lake: From July to August 2020, persistent rainstorms in the middle and lower reaches of the Yangtze River also caused serious floods in Poyang Lake, Jiangxi Province, China. Poyang Lake, located between $115°28' \sim 116°27'E$ and $28°13' \sim 29°28'N$, has similar geographic and hydrological conditions to Dongting Lake. To analyze the flood event in Poyang Lake, 20 dual-polarized SAR images from February to September 2020 were utilized, of which the images from February to June 2020 were selected to form the multitemporal reference dataset, and the image on August 19, 2020, was selected to generate the flood map.

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TABLE VI
PARAMETER SETTING IN DIFFERENT FLOOD-MAPPING REGIONS

Target objects	Poyang Lake(a/b)	Peru (a/b)
Temporarily Submerged Area	0.2/0.7	0.5/0.8
Permanent Water	0.1/0.2	0.1/0.4
Permanent Land	0.1/0.3	0.1/0.3

2) Lago La Nina: In mid-March 2017, a large-scale flood occurred in the rural area of Parachique, Peru, affecting areas of approximately 33 250 hectares. Within flooded areas, the Lago La Nina (lake region) was selected for the study. It covers land from $80°24'$ to $80°51'W$ and $5°31'$ to $5°51'S$ in the southern hemisphere. The dataset was built by using 18 dual-polarized SAR images available from February to October 2017. Among them, the images from June to October 2017 were regarded as multitemporal references of nonflood states. The image on March 23, 2017, was employed for flood mapping.

Fig. 9 gives the flood-mapping results of the proposed method in two other flood cases near lake regions. In line with the performance in East Dongting Lake, VMSTF well distinguishes flood areas from PEW bodies and highlights the regions of half-submerged vegetation. Since the days we used to map the floods were separately at the early stage and the peak flooding stage of Lago La Nina and Poyang Lake flood events, larger regions of FSV were expected to appear around Lago La Nina than Poyang Lake. This expectation is consistent with the actual mapping results displayed in Fig. 9. From the spatial distribution of flood-mapping results, Poyang Lake shows high similarity to Dongting Lake since they possess similar climatic conditions, geographical locations, and landforms around the lake. Besides flood-mapping results, the spatiotemporal flood characteristics derived from synthetic six-class attribute maps can also help to analyze the whole flood event. For example, when pixels are marked with "PEL" in time series and "Water" in space, it often indicates serious floods. Moreover, when pixels are marked with "TSA" in the time series and "Nonwater" in space, it is possible that the flood has not spread to the area or has receded at the mapping time (before FSV activation).

Affected by different climates and geographical environments, the temporal and spatial characteristics of flood events vary slightly. To ensure a reliable mapping result, the parameters in the multitemporal feature extraction need to be adjusted appropriately. Generally speaking, the determination of the parameters in fuzzy membership functions can be realized based on the histogram distribution of the multitemporal analysis results, as introduced aforementioned in Fig. 4. Fortunately, the parameters involved in this method are insensitive as we analyzed above. To simplify the parameter adjustment, the parameters in Poyang Lake were set the same as they were used in East Dongting Lake, and the Lago La Nina was tuned slightly as shown in Table VI.

C. Transferability Across Flood Time

Due to the accumulation of sufficient SAR data over these years, the VMSTF method can accurately map the flooded area by making full use of information from both spatial and temporal domains. However, considering the interannual

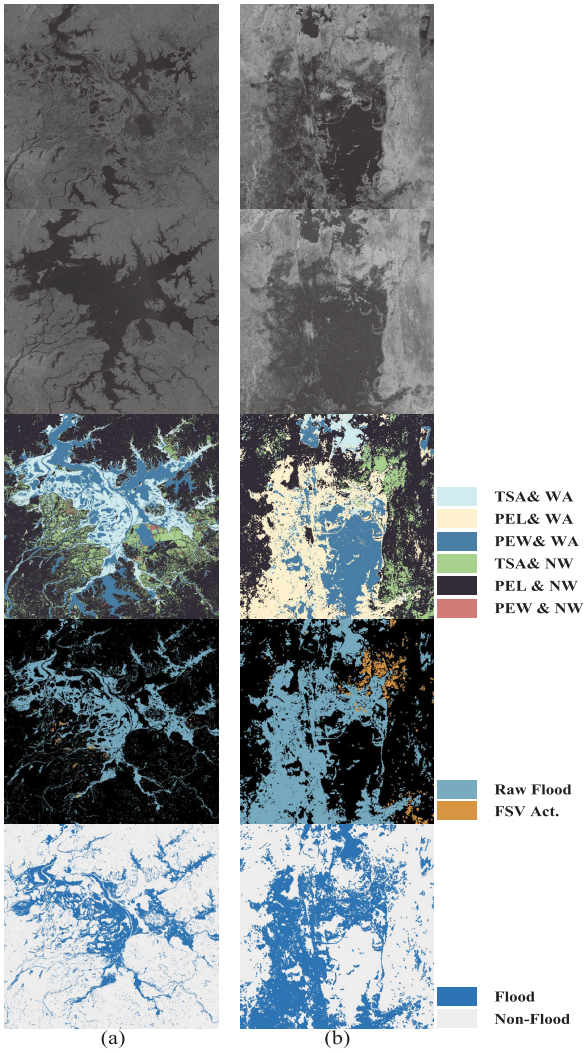


Fig. 9. Flood-mapping performance of VMSTF across geographical locations. From top to bottom: Sentinel-1 SAR images before flood events, Sentinel-1 SAR images on a flood day, synthetic six-class attribute maps, FSV activation, and final flood map. (a) Poyang Lake, China. (b) Lago La Nina, Peru.

similarity of flood events in the same region, whether existing/acquired multitemporal datasets can be flexibly applied in flood-mapping tasks for other years becomes an interesting and meaningful question. To verify the reusability of the multitemporal dataset in the proposed framework, we selected one day each year from the rainy season of East Dongting Lake from 2017 to 2019 to conduct flood-mapping experiments. Results of VMSTF are shown in Fig. 10.

When reusing the multitemporal dataset in 2020, the average OA of flood extraction results in other test years is 94.73% (The OA values for 2017, 2018, and 2019 are 0.9584, 0.9397, and 0.9438, respectively). This value, just slightly lower than the flood extraction accuracy in 2020 (see Table III), demonstrates that the proposed flood-mapping framework possesses good "compatibility" with the multitemporal dataset. It is easily understood because the multitemporal analysis is based on a complete change cycle in time series and can well reflect the inherent differences among different objects. Namely, in the case of limited or lack of data, the existing–historical multitemporal dataset can be used to

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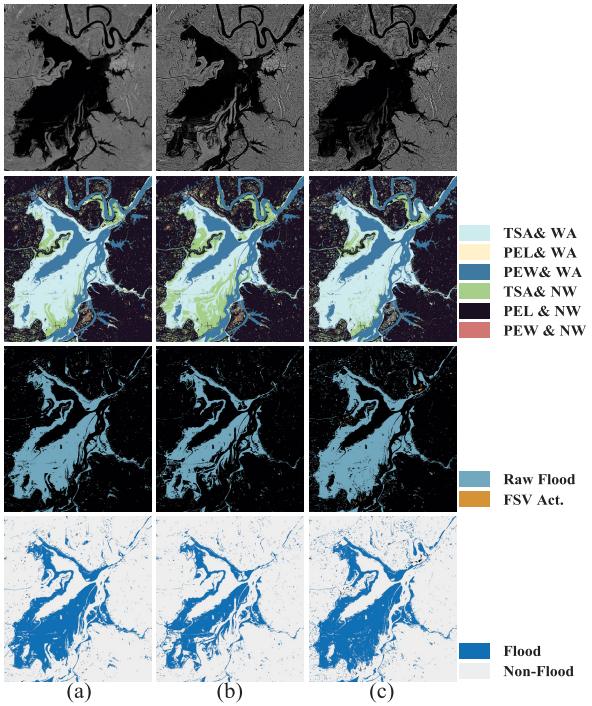


Fig. 10. Flood-mapping performance of VMSTF in East Dongting Lake across time. From top to bottom: Sentinel-1 SAR images on flood days, synthetic six-class attribute maps, FSV activation, and final flood map. (a) July 17, 2017. (b) August 17, 2018. and (c) July 19, 2019.

evaluate the current flood event, broadening the application scenarios of the proposed methods. When comparing the flood extraction accuracy between different years, it can be found that the result in 2018 is slightly worse. This is because the flood events in 2017 and 2019 are similar to those in 2020, while no floods occurred in 2018. Fortunately, the extraction accuracy of VMSTF is still satisfactory even if there are obvious differences in temporal changing characteristics between 2018 and 2020, which once again verifies the stability and effectiveness of the proposed method. Furthermore, VMSTF can capture the differences between flood events in different years by using the multitemporal dataset of a certain year as the benchmark, which provides support for the interannual comparative analysis of flood events.

V. Conclusion

Rapid and accurate flood mapping is of great importance for emergency response. Aiming at this target, we have proposed a novel unsupervised VMSTF flood-mapping method by using dual-polarized Sentinel-1 GRD products in this work. Based on the analyses of time-series variation in backscatter values for four typical flood ground objects over a hydrological year, flood features in temporal and spatial domains are first characterized under appropriate polarization modes by variation modeling and fuzzy-based methods. Then, the proposed framework fuses spatiotemporal features and performs an FSV activation for flood mapping. The experiments undertaken on the 2020 flood event of East Dongting Lake confirmed that the proposed VMSTF method can display better flood-mapping performance in both qualitative and quantitative assessments. Extensive experiments on parameter sensitivity, data uncertainty, and model transferability across different times and geographical locations further demonstrate the robustness and universality of the proposed framework.

However, there is still room for improvement in practical applications. On the one hand, the proposed method requires a certain number of nonflood SAR images at the same monitoring location for reference, which may cause difficulties in data preparation. On the other hand, due to the complexity of ground objects in flood areas, slight adjustments to the fuzzy rule parameters for the variation model are still necessary. To further ease the use and upgrade the universality of the proposed method, we plan to improve the variation model to better capture the flood features in the temporal domain with fewer references and investigate the feasibility of using nonflood SAR images of other locations as an alternative. Moreover, it is also a promising future research direction to broaden the flood-mapping framework under different flood scenarios, such as plains, rivers, and lakes, by incorporating the relevant physical models on flood propagation mechanisms and topographic characteristics.

References

[1] S. Hallegatte, C. Green, R. J. Nicholls, and J. Corfee-Morlot, "Future flood losses in major coastal cities," *Nature Climate Change*, vol. 3, no. 9, pp. 802–806, Sep. 2013.

[2] M. Huang and S. Jin, "Rapid flood mapping and evaluation with a supervised classifier and change detection in Shouguang using Sentinel-1 SAR and Sentinel-2 optical data," *Remote Sens.*, vol. 12, no. 13, p. 2073, Jun. 2020.

[3] M. Rättich, S. Martinis, and M. Wieland, "Automatic flood duration estimation based on multi-sensor satellite data," *Remote Sens.*, vol. 12, no. 4, p. 643, Feb. 2020.

[4] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[5] Z. Dong, G. Wang, S. O. Y. Amankwah, X. Wei, Y. Hu, and A. Feng, "Monitoring the summer flooding in the Poyang Lake area of China in 2020 based on Sentinel-1 data and multiple convolutional neural networks," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 102, Oct. 2021, Art. no. 102400.

[6] M. Wood, S. M. de Jong, and M. W. Straatsma, "Locating flood embankments using SAR time series: A proof of concept," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 70, pp. 72–83, Aug. 2018.

[7] M. Shangguan, W. Wang, and S. Jin, "Variability of temperature and ozone in the upper troposphere and lower stratosphere from multi-satellite observations and reanalysis data," *Atmos. Chem. Phys.*, vol. 19, no. 10, pp. 6659–6679, May 2019.

[8] A. Ahamed and J. D. Bolten, "A MODIS-based automated flood monitoring system for Southeast Asia," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 61, pp. 104–117, Sep. 2017.

[9] X. Shen, D. Wang, K. Mao, E. Anagnostou, and Y. Hong, "Inundation extent mapping by synthetic aperture radar: A review," *Remote Sens.*, vol. 11, no. 7, p. 879, Apr. 2019.

[10] D. O'Grady, M. Leblanc, and D. Gillieson, "Use of ENVISAT ASAR global monitoring mode to complement optical data in the mapping of rapid broad-scale flooding in Pakistan," *Hydrol. Earth Syst. Sci.*, vol. 15, no. 11, pp. 3475–3494, Nov. 2011.

[11] D. C. Mason, I. J. Davenport, J. C. Neal, G. J.-P. Schumann, and P. D. Bates, "Near real-time flood detection in urban and rural areas using high-resolution synthetic aperture radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 8, pp. 3041–3052, Aug. 2012.

[12] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst. Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.

[13] J. Kittler and J. Illingworth, "Minimum error thresholding," *Pattern Recognit.*, vol. 19, no. 1, pp. 41–47, Jan. 1986.

[14] J. Liang and D. Liu, "A local thresholding approach to flood water delineation using Sentinel-1 SAR imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 53–62, Jan. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:42:25 UTC from IEEE Xplore. Restrictions apply.

[15] A. Likas, N. Vlassis, and J. J. Verbeek, "The global k-means clustering algorithm," *Pattern Recognit.*, vol. 36, no. 2, pp. 451–461, Feb. 2003.

[16] W. Peizhuang, "Pattern recognition with fuzzy objective function algorithms (James C. Bezdek)," *SIAM Rev.*, vol. 25, no. 3, p. 442, Jul. 1983.

[17] R. Xu and D. Wunsch II, "Survey of clustering algorithms," *IEEE Trans. Neural Netw.*, vol. 16, no. 3, pp. 645–678, May 2005.

[18] G. Boni et al., "A prototype system for flood monitoring based on flood forecast combined with COSMO-SkyMed and Sentinel-1 data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2794–2805, Jun. 2016.

[19] R. Metrikasari et al., "Mapping of flood prone area in Jakarta using fuzzy C-means," in *Proc. Int. Conf. Data Sci. Appl. (ICoDSA)*, Oct. 2021, pp. 157–162.

[20] W. Cao, S. Martinis, and S. Plank, "Automatic SAR-based flood detection using hierarchical tile-ranking thresholding and fuzzy logic," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5697–5700.

[21] F. Polverari, S. Mori, N. Pierdicca, F. S. Marzano, and L. Pulvirenti, "Precipitation signature on side-looking aperture radar imaging: Sensitivity analysis to surface effects at C, X and Ku band," in *Proc. 11th Eur. Radar Conf.*, Oct. 2014, pp. 197–200.

[22] H. C. Jung et al., "Characterization of complex fluvial systems using remote sensing of spatial and temporal water level variations in the Amazon, Congo, and Brahmaputra Rivers," *Earth Surf. Processes Landforms*, vol. 35, no. 3, pp. 294–304, Mar. 2010.

[23] L. Pulvirenti, M. Chini, N. Pierdicca, and G. Boni, "Use of SAR data for detecting floodwater in urban and agricultural areas: The role of the interferometric coherence," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1532–1544, Mar. 2016.

[24] I. Hajnsek, T. Jagdhuber, H. Schon, and K. P. Papathanassiou, "Potential of estimating soil moisture under vegetation cover by means of PolSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 2, pp. 442–454, Feb. 2009.

[25] G. Moser and S. B. Serpico, "Generalized minimum-error thresholding for unsupervised change detection from SAR amplitude imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 10, pp. 2972–2982, Oct. 2006.

[26] J. Lu, L. Giustarini, B. Xiong, L. Zhao, Y. Jiang, and G. Kuang, "Automated flood detection with improved robustness and efficiency using multi-temporal SAR data," *Remote Sens. Lett.*, vol. 5, no. 3, pp. 240–248, Mar. 2014.

[27] Y. Li, S. Martinis, S. Plank, and R. Ludwig, "An automatic change detection approach for rapid flood mapping in Sentinel-1 SAR data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 73, pp. 123–135, Dec. 2018.

[28] B. Yang, L. Qin, J. Liu, and X. Liu, "UTRNet: An unsupervised time-distance-guided convolutional recurrent network for change detection in irregularly collected images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410516.

[29] L. Giustarini, R. Hostache, P. Matgen, G. J.-P. Schumann, P. D. Bates, and D. C. Mason, "A change detection approach to flood mapping in urban areas using TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 4, pp. 2417–2430, Apr. 2013.

[30] D. Amitrano, G. Di Martino, A. Iodice, D. Riccio, and G. Ruello, "Unsupervised rapid flood mapping using Sentinel-1 GRD SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3290–3299, Jun. 2018.

[31] P. Matgen, R. Hostache, G. Schumann, L. Pfister, L. Hoffmann, and H. H. G. Savenije, "Towards an automated SAR-based flood monitoring system: Lessons learned from two case studies," *Phys. Chem. Earth, A/B/C*, vol. 36, nos. 7–8, pp. 241–252, Jan. 2011.

[32] M. Chini, R. Hostache, L. Giustarini, and P. Matgen, "A hierarchical split-based approach for parametric thresholding of SAR images: Flood inundation as a test case," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 6975–6988, Dec. 2017.

[33] L. Landuyt, A. Van Wesemael, G. J.-P. Schumann, R. Hostache, N. E. C. Verhoest, and F. M. B. Van Coillie, "Flood mapping based on synthetic aperture radar: An assessment of established approaches," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 722–739, Feb. 2019.

[34] S. Martinis, J. Kersten, and A. Twele, "A fully automated TerraSAR-X based flood service," *ISPRS J. Photogramm. Remote Sens.*, vol. 104, pp. 203–212, Jun. 2015.

[35] X. Zhang, N. W. Chan, B. Pan, X. Ge, and H. Yang, "Mapping flood by the object-based method using backscattering coefficient and interference coherence of Sentinel-1 time series," *Sci. Total Environ.*, vol. 794, Nov. 2021, Art. no. 148388.

[36] F. Cian, M. Marconcini, and P. Ceccato, "Normalized difference flood index for rapid flood mapping: Taking advantage of EO big data," *Remote Sens. Environ.*, vol. 209, pp. 712–730, May 2018.

[37] J. Zhao et al., "Deriving exclusion maps from C-band SAR time-series in support of floodwater mapping," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112668.

[38] B. DeVries, C. Huang, J. Armston, W. Huang, J. W. Jones, and M. W. Lang, "Rapid and robust monitoring of flood events using Sentinel-1 and Landsat data on the Google Earth engine," *Remote Sens. Environ.*, vol. 240, Apr. 2020, Art. no. 111664.

[39] X. Shen, E. N. Anagnostou, G. H. Allen, G. Robert Brakenridge, and A. J. Kettner, "Near-real-time non-obstructed flood inundation mapping using synthetic aperture radar," *Remote Sens. Environ.*, vol. 221, pp. 302–315, Feb. 2019.

[40] M. Singha et al., "Identifying floods and flood-affected paddy Rice fields in Bangladesh based on Sentinel-1 imagery and Google Earth engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 278–293, Aug. 2020.

[41] W. Wu, W. Wang, M. E. Meadows, X. Yao, and W. Peng, "Cloud-based typhoon-derived paddy Rice flooding and lodging detection using multi-temporal Sentinel-1&2," *Frontiers Earth Sci.*, vol. 13, no. 4, pp. 682–694, Dec. 2019.

[42] K. M. Brown, C. H. Hambidge, and J. M. Brownett, "Progress in operational flood mapping using satellite synthetic aperture radar (SAR) and airborne light detection and ranging (LiDAR) data," *Prog. Phys. Geography, Earth Environ.*, vol. 40, no. 2, pp. 196–214, Apr. 2016.

[43] N. Pierdicca, L. Pulvirenti, and M. Chini, "Flood mapping in vegetated and urban areas and other challenges: Models and methods," in *Flood Monitoring Through Remote Sensing*. Cham, Switzerland: Springer, 2018, pp. 135–179.

[44] N. Engheta and C. Elachi, "Radar scattering from a diffuse vegetation layer over a smooth surface," *IEEE Trans. Geosci. Remote Sens.*, vol. GE-20, no. 2, pp. 212–216, Apr. 1982.

[45] J. Richards, G.-Q. Sun, and D. Simonett, "L-band radar backscatter modeling of forest stands," *IEEE Trans. Geosci. Remote Sens.*, vol. GE-25, no. 4, pp. 487–498, Jul. 1987.

[46] M. A. Clement, C. G. Kilsby, and P. Moore, "Multi-temporal synthetic aperture radar flood mapping using change detection," *J. Flood Risk Manage.*, vol. 11, no. 2, pp. 152–168, Jun. 2018.

[47] A. Twele, W. Cao, S. Plank, and S. Martinis, "Sentinel-1-based flood mapping: A fully automated processing chain," *Int. J. Remote Sens.*, vol. 37, no. 13, pp. 2990–3004, Jul. 2016.

[48] T. Lei, X. Jia, Y. Zhang, S. Liu, H. Meng, and A. K. Nandi, "Superpixel-based fast fuzzy C-means clustering for color image segmentation," *IEEE Trans. Fuzzy Syst.*, vol. 27, no. 9, pp. 1753–1766, Sep. 2019.

[49] M. Gong, Z. Zhou, and J. Ma, "Change detection in synthetic aperture radar images based on image fusion and fuzzy clustering," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2141–2151, Apr. 2012.

[50] J. C. Bezdek, *Pattern Recognition With Fuzzy Objective Function Algorithms*. Cham, Switzerland: Springer, 2013.

[51] J. J. Saade and H. B. Diab, "Defuzzification techniques for fuzzy controllers," *IEEE Trans. Syst. Man Cybern., B, Cybern.*, vol. 30, no. 1, pp. 223–229, 2000.

[52] G. Guo, H. Wang, D. Bell, Y. Bi, and K. Greer, "KNN model-based approach in classification," in *Proc. OTM Confederated Int. Conf. Move Meaningful Internet Syst.* Cham, Switzerland: Springer, 2003, pp. 986–996.

[53] P. Gong et al., "Stable classification with limited sample: Transferring a 30-m resolution sample set collected in 2015 to mapping 10-m resolution global land cover in 2017," *Sci. Bull.*, vol. 64, no. 6, pp. 370–373, Mar. 2019.

[54] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[55] Y. Xu, B. Du, F. Zhang, and L. Zhang, "Hyperspectral image classification via a random patches network," *ISPRS J. Photogramm. Remote Sens.*, vol. 142, pp. 344–357, Aug. 2018.

[56] J. Gawlikowski et al., "A survey of uncertainty in deep neural networks," *Artif. Intell. Rev.*, pp. 1–77, Jul. 2023, doi: 10.1007/s10462-023-10562-9.

[57] S. K. McFeeters, "The use of the normalized difference water index (NDWI) in the delineation of open water features," *Int. J. Remote Sens.*, vol. 17, no. 7, pp. 1425–1432, May 1996.

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